SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Short Answers

Course Code:	ITA03 01	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – Short Answers
Weightage:	30 % of CIA	**Total Marks:**	100 Marks
CO1 Statement:	*Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
Student Name:	G. Anjani Kumar	**Reg. No.:**	192373022
Section / Batch:	CSE-DS / 2023	**Date:**	09/10/2026

Code of Conduct

I, G. Anjani Kumar 192373022 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: G. Anjani Kumar

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

Faculty In-charge	Course Coordinator	Head of Department
Dr. J Gnanavel		
Signature:	Signature:	Signature:
Date: 09/08/26	Date:	Date:

96/100

Short Answers Test — Mobile Computing

G. Arjun Kumar.
19231302
CSE-DS

Given

1. 45 MHz - Spectrum auction
   25 KHz - Simplex Channel

   45 MHz = 45,000

   45000 KHz / 2×25 = 900

   900 Channels.

2. 500 total voice channels
   7 Cell frequency

   $\frac{500}{7} = 80$

3. 800 simultaneous voice channels.

   100 × 8 = 800 Channels.

4. radiates = 80W
   transmits = 4W

   $10 \times \log_{10}(80/4) = 10 \times \log_{10}(20)$

   = 13.01 dB.

5. Signal at = -60dB

   Co-Channel interference at = -92dBm

   -60dBm - (-92dBm) = 32dB

   -60 + 92 = 32dB

6. 42 MHz, 300kHz

+

42MHz

42,000 KHz, 300KHz, 1Temperature

$\frac{42000}{300} = 140$

$\frac{140}{7}$ = 20 channels.

7. 64 - receiver, 4 - Path loss exponent.

$\frac{64\,W}{2^4} = \frac{64}{16} = 4W$

8. By a factor of 4 $\Rightarrow 2^2 = 4$.

9. Cluster size = 4

15 voice channels.

4 X 15 = 60 Channels.

10. vehicle traveling at 100 km/h

Carrier 900 MHz

$\frac{27.78\,m/s}{3 \times 10^8} \times 900 \times 10^6\,Hz = 83.33$

11.

$\frac{5}{8}\,ms = 0.625\,ms.$

1200KHz

$\frac{24000}{200} = 120$ carriers

13.

SW = 39 dBm

39 dBm − 33 = 6dBm

14.

$20 \log 10(2500) + 20 \log 10(1900 \times 10^6) - 147.55$

$= 105.99 dB$

15. 6 cells, 250 calls

$6 \times 250 = 1500$ calls

16. 4000 new call 3mins/min

4000 calls $\times$ 180 $= 720,000$ calls.

17. 150 channels, 25KHz

$150 \times 25 KHz = 3750 KHz$

$= 3.75 MHz$

18. 29 "internal guard, 40KHz

$29 \times 40 = 1.16 MHz$

19. 90 km/h, 3 km per cell

$\frac{90}{3}$ = 30 hands off.

20 8 regional, 40 local stations, 250 users

8 × 40 × 250 = 80,000 users

21. 400 sq km, 400 m radius

$\frac{400\ km^2}{(0.4)^2} = \frac{400}{0.16}$ = 2500 cells.

22. 30 kHz each channel

6 MHz - spectrum pool.

$\frac{6,000\ kHz}{30\ kHz}$ = 200 channels

23. 1200 subscribers 4 calls per hour, 2.5 minutes per call.

1200 × 4 × (2.5/60) = 200 Erlangs.

24. 2.5 GHz, 40 MHz bandwidth

$\frac{40,000}{400}$ = 100 users.

25. 600 users, 1200 stations

= 1200 × 600

= 720,000 users.

$\frac{200}{4}$ = 50 trunks

28. 100,000 connections
40,000 call attempts
3 mins

$\frac{120,000 - 100,000}{120000} = \frac{20000}{120000} = 16.67\%$

21 15%
600,000 subscribers

$= 600,000 \times 0.15$

$= 90,000$ subscribers.

29. 12,000 calls/s . 9,000 calls/s

$\frac{9000}{12000} = 0.75$

$0.75 \times 100 = 75\%$

30. factor 4 across, 48MHz, 200 KHz per channel.

$\frac{48,000}{200} = 240$

$\frac{240}{4} = 60$ channels

31. 900 MHz, 200 kHz uses, 30 MHz.

$\frac{30,000}{200}$ = 150 channels.

32. 2500 handoff, 99%

2500 × 0.99 = 2475 calls.

33.

12W ≈ 40.79 dBm

40.79 dBm − 45 dBm

= −4.21 dBm

34. 7-cell cluster, 900 users.

7 × 900

= 6300 users.

35. 8 equal slots.

4.8 ms / 8

$\frac{4.8}{8}$ = 0.6 ms

36. 150 km/h, 1.8 GHz

$\frac{41.67 \text{ m/s}}{3 \times 10^8}$ × 1.8×10^9 = 250 Hz

30,000 Calls

10,000/min , 4min duration.

$$\frac{280,000 - 250,000}{280,000} = \frac{30,000}{280,000} = 10.71\%$$

38 9 cells

600 users.

9 × 600 = 5400 users

39. 60,000 calls/s

0.4%.

= 60000 × 0.004

= 240 cells.

40 1800 MHz , 43 dBm , 2km

43 - (-87) = 130 dB.

41. 2.5 km^2 , 150 identical stations.

= 150 × 2.5 km^2

= 375 km^2

1.

$$\frac{48000}{200} = \frac{200}{6} = 40 \text{ channels.}$$

43. 6000 users, 150,000 users
$\frac{150000}{6000} = 25$ cells

44. 9 mins, 3 mins.
9 - 3 = 6 mins

45. -85 dBM, -115 dBM
-85 - (-115) = -85 + 115
= 30 dB.

46. 250 users, 150 KBPS
= 250 × 150
= 37,500 KBPS
= 37.5 MBPS

49. 1500 subscribers, 2.5 mins
1500 × 4 × (2.5/60) = 250 exchanges.

48. 250, 12 KHz
= 250 × 12 KHz
= 3,000 KHz
= 3 MHz

47. 1300 connection
600 KBPS
= 1300 × 600
= 900,000 KBPS
= 900 MBPS.

50. 4 KM, 350 KM² area = πr²
$\frac{350}{4\pi} = \frac{350}{50.27}$ = 6.96 towers.